

ETX-R41K111C

EtherCAT Controller User Manual

Version: V1.0 2025 Feb 26

To properly use the product, read this manual thoroughly is necessary.

Revision History

Date	Revision	Description
2025/02/26	1.0	Document creation.

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Electrical safely

- To prevent electrical shock hazard, disconnect the power cable from the electrical outlet before relocating the system.
- When adding or removing devices to or from the system, ensure that the power cables for the devices are unplugged before the signal cables are connected. Disconnect all power cables from the existing system before you add a device.
- Before connecting or removing signal cables from motherboard, ensure that all power cables are unplugged.
- Seek professional assistance before using an adapter or extension card. These devices could interrupt the grounding circuit.
- Make sure that your power supply is set to the voltage available in your area.
- If the power supply is broken, contact a qualified service technician or your retailer.

Operational safely

- Please carefully read all the manuals that came with the package, before installing the new device.
- Before use ensure all cables are correctly connected and the power cables are not damaged. If you detect and damage, contact the dealer immediately.
- To avoid short circuits, keep paper clips, screws, and staples away from connectors, slots, sockets and circuitry.
- Avoid dust, humidity, and temperature extremes. Do not place the product in any area where it may become wet.
- If you encounter technical problems with the product, contact a qualified service technician or the dealer.

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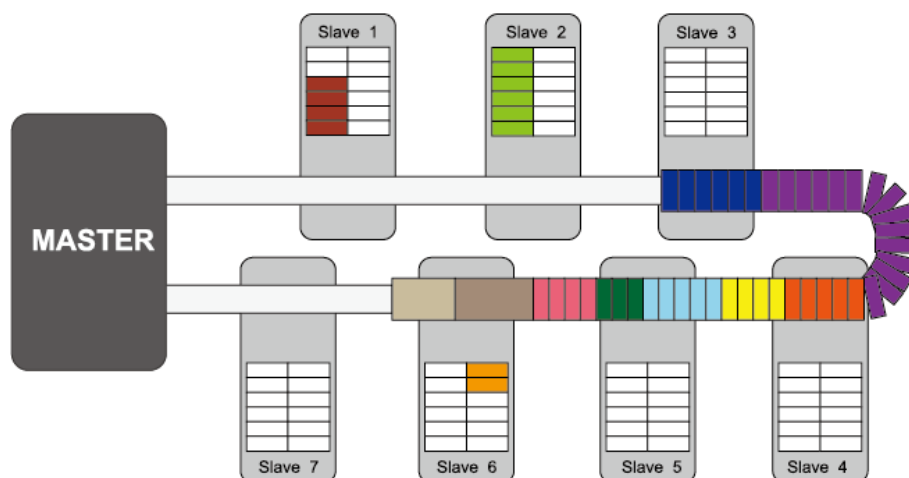
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1. Introduction

1.1. Overview



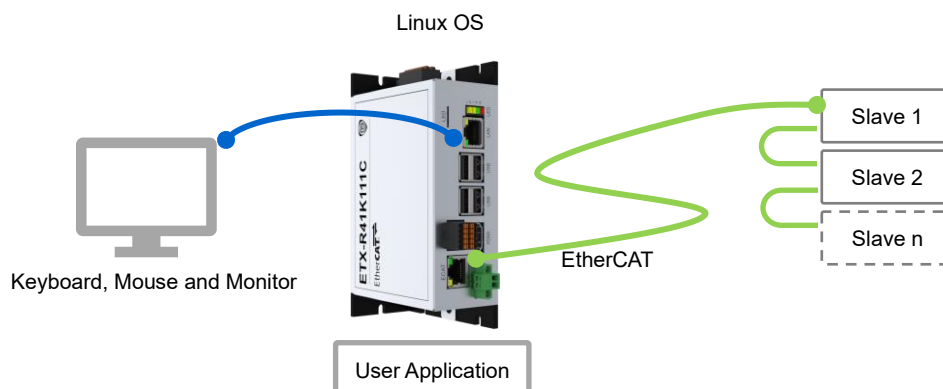
ETX-R41K111C is an EtherCAT Master Controller powered by Raspberry Pi. It provides an efficient and flexible solution for motion control in automation applications. By utilizing the computational power and expandability of Raspberry Pi, along with the high-speed and real-time EtherCAT communication protocol, it enables precise and rapid motion control.



ETX-R41K111C offers a dual-scenario operation, making it a versatile and flexible solution for motion control in automation applications.

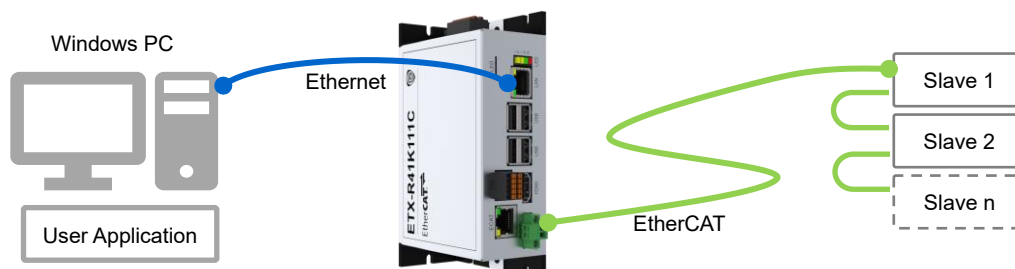
Scenario 1:

Standalone configuration for direct control as an independent EtherCAT controller, without needing an external host. By adopting this configuration, all applications run directly on Linux, with development possible on either Windows or Linux platforms.

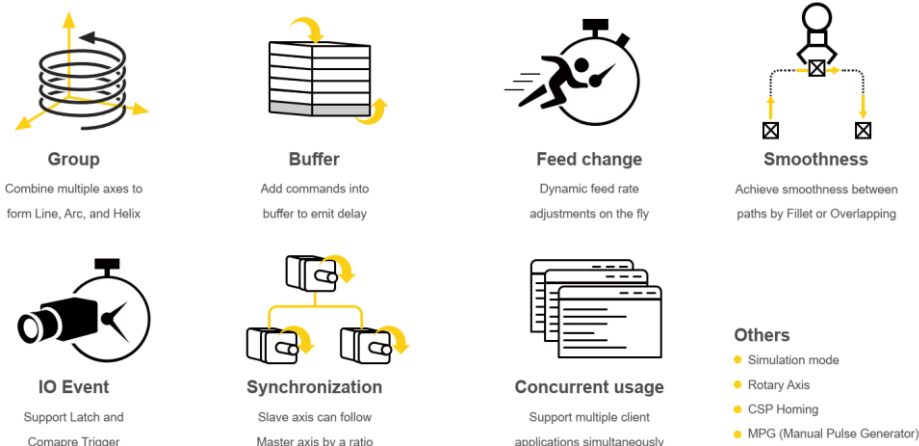


Scenario 2:

Control via Ethernet using EtherCAT communication for automation systems. This configuration is more straightforward for Windows users, as it allows for easier integration with common development tools and software environments typically found in Windows-based systems.



Regardless of whether choosing the Ethernet or Standalone configuration, both options can effectively execute excellent EtherCAT motion control.



1.2. Specification

ETX-R41K111C is not only an EtherCAT master solution but also features advanced I/O implemented by FPGA, allowing virtually instantaneous operation without any time delay, ensuring precise position control and measurement.

ETX-R41K111C - EtherCAT Controller acting as Motion and I/O Master with ENC and CMP

■ Hardware

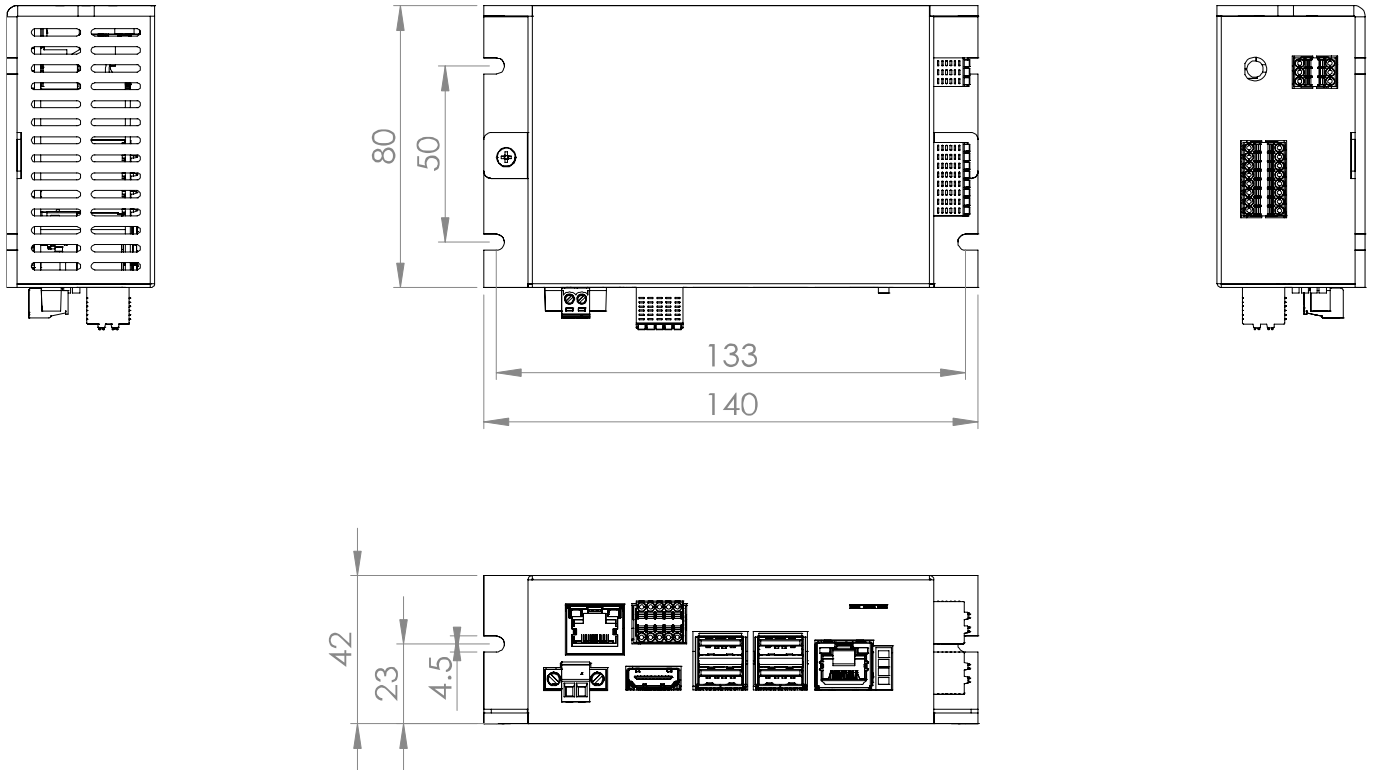
EtherCAT	
Axes and IO	Max. 36 Axes, and 100 DIO Slaves Support 6 / 12 / 18 / 24 / 30 / 36 Axes
Distributed Clocks	0.5 or 1 ms
CoE	Supports read / write of PDO and SDO objects
CiA402	Supports PP, CSP and Homing mode.
Development	Supports C/C++ or C#, for 32/64-bit application
Operation System	Scenario 1 - Raspberry Pi OS Scenario 2 - Windows 7 or Windows 10/11 LTSC
Utility tool	ECATNavi: Test and diagnosis EtherCAT ECATScan: Generate and modify ENI files.
Isolated IO	
Digital Input	4-CH, NPN/PNP
Digital Output	3-CH, 250 mA per channel
ENC and CMP (ETX-R41K111C)	
Encoder Input.	2-CH, 32-bit EA/EB
Encoder Input Frequency	Max. 4MHz, for CW/CCW and PULSE/DIR Mode Max. 9.8MHz for 4xAB Phase Mode
Latch Input	2-CH, 1024pt per channel, Max. 10KHz
Trigger Output	2-CH, Differential Line Driver
	Auto Trigger, Max. trigger frequency 1MHz, 1 ~ unlimited counts
	Table Trigger, 4096pts allocatable for all channels
	Toggle Trigger, 4096pts allocatable for all channels
Trigger Source	Selectable and Repeatable from Encoder
Trigger Pulse Width	Programmable, 0.1 ~ 6553.5 or 1 ~ 65535 micro-second
System	
SoC	ARM Broadcom 4-core

RAM	4GB
Storage	eMMC 32GB
USB	4 x USB Type-A(Max. total current 900mA)
Ethernet	1 x Gigabit Ethernet
Display	1 x HDMI
Serial	2 x RS-485
General	
Power Input	DC 24V
Power consumption	5W typical no load
Working temperature	0°C~50°C (32°F~122°F) ambient temperature with air flow
Storage temperature	-20°C~80°C (-4°F~176°F)
Humidity	85% (non-condensing @60°C)
Dimension	L140 x W80 x H42 mm

■ Software – EtherCAT Motion Library on ETX-R41K111C

Function		Scenario 1	Scenario 2
Commend Mode	Sync and Async	☐	☐
Home Mode	CSP Homing	☐	☐
Motion Profile	T-Curve and S-Curve	☐	☐
Group for Interpolation	Linear (All axes), 2D/3D Arc, Helical and Cone	☐	☐
Command Buffer	1,000 for all	☐	☐
Change on the Fly	Target Position and Velocity	☐	☐
Path Smoothing	Fillet and Overlap	☐	☐
Soft Limit	Positive and Negative	☐	☐
Ring Counter	Rotary Axis	☐	☐
Event Trigger	Realtime Latch and Comparator	☐	☐
Manual Pulse Generator	Movement Follow MPG	☐	☐
E-Gear	Rotating Cut	☐	☐
E-Gear	Flying Cut	☐	☐
E-Gear	Gantry	☐	☐
Simulation Mode	Execute without Slave Module	☐	☐
Ethernet Mode	Execute from Ethernet	☐	☐

1.3. Dimension

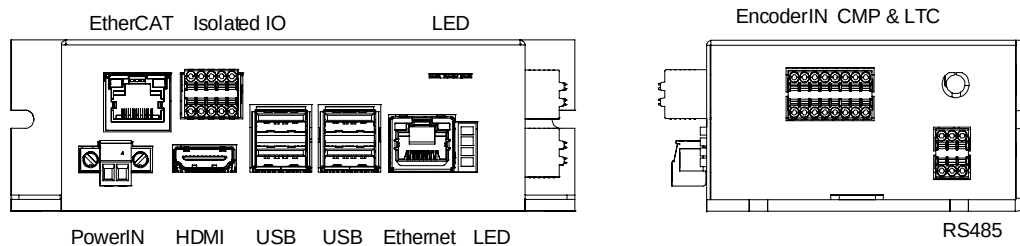


unit: mm

1.4. Packing List

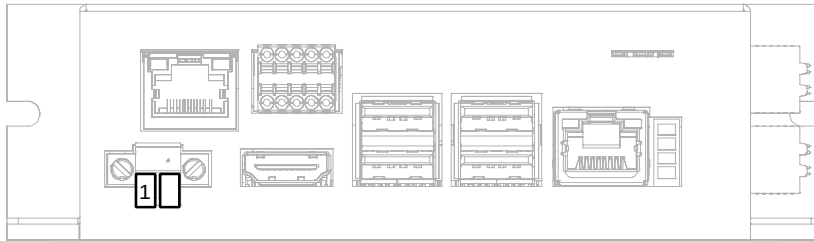
Item	Description
	ETX-R41K111C EtherCAT Motion Controller

1.5. Placement



Label	Description
DC IN	Power Input Connector
HDMI	HDMI
USB	USB Ports
Ethernet	Ethernet Communication Port
EtherCAT	EtherCAT Master Communication Port
Isolated IO	Digital Input and Output Connector
Encoder IN	Encoder Input, CMP & LTC Connector
RS485	RS485 Ports

1.5.1. Power Input Connector

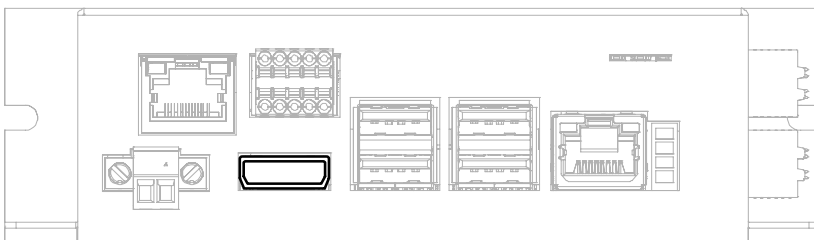


Pin	Label	Description
1	+	24V DC Power input
2	-	24V DC Ground

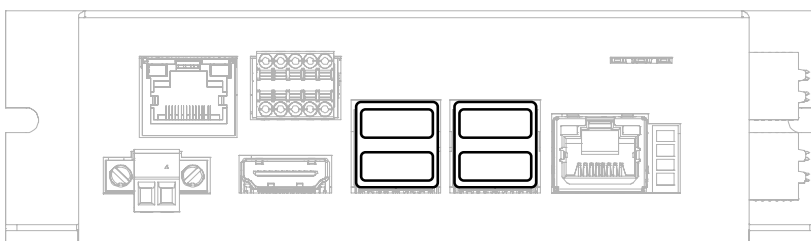
Note

- Upon power-up, the controller will boot automatically.
- DO NOT connect to AC power.
- Be sure to connect the positive and negative terminals of the power supply correctly.

1.5.2. HDMI



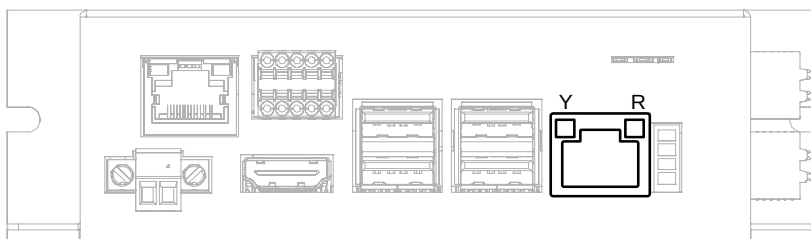
1.5.3. Raspberry Pi USB Ports



Note

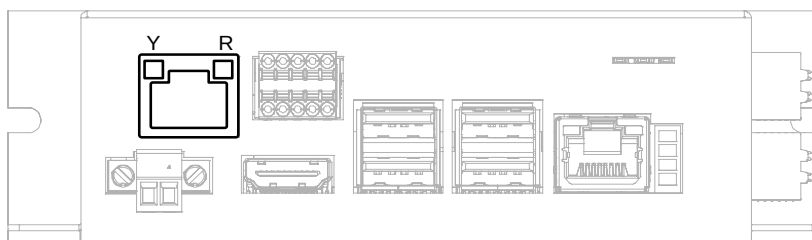
- All USB ports are shared through a USB hub from the host,

1.5.4. Raspberry Pi Ethernet Communication Port



LED	Description
Left (Yellow)	On – Operating is connection. Off – Operating is not connection.
Right (Green)	Blinking – Ethernet data is transmitting or receiving Off – EtherCAT is not established.

1.5.5. EtherCAT Master Port

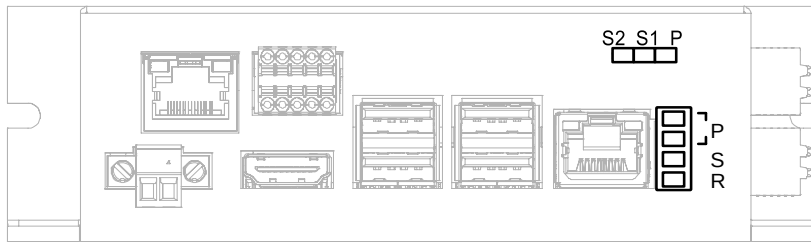


LED	Description
Left (Yellow)	On – No slave is connected physically. Off – Slave is connected physically.
Right (Green)	Blinking – EtherCAT data is transmitting or receiving Off – EtherCAT is not established.

Note

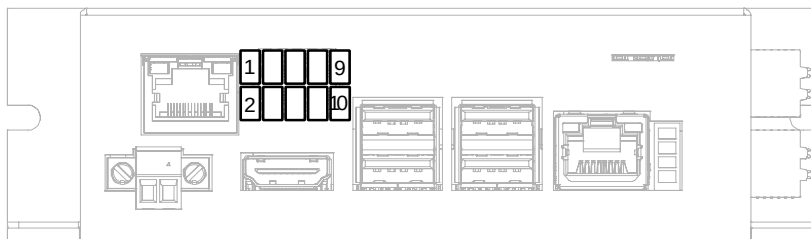
- Use at least CAT5e network cable for 100Mbps communication.

1.5.6. LED Indicators



LED	Color	Behavior
P	Yellow	Power On.
S	Green	System accessing.
R	Red	System On.
STS1	Green	Blinking, Ethernet Connection Ready for S2
STS2	Green	Reserved
P	Yellow	Power On.

1.5.7. Digital Input and Output Connector

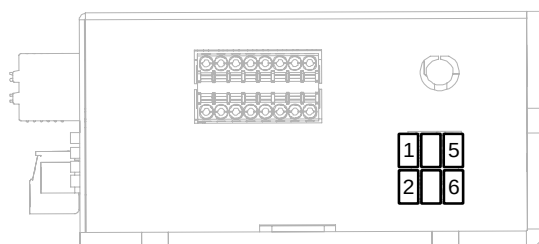


Pin	Label	Description	Pin	Label	Description
1	DI1	Digital Input 1	2	DO1	Digital Output 1
3	DI2	Digital Input 2	4	DO2	Digital Output 2
5	DI3	Digital Input 3	6	DO3	Digital Output 3
7	DI4	Digital Input 4	8	COM	Common Contact
9	E24V	DC 24V Input	10	EGND	DC 24V Ground

Note

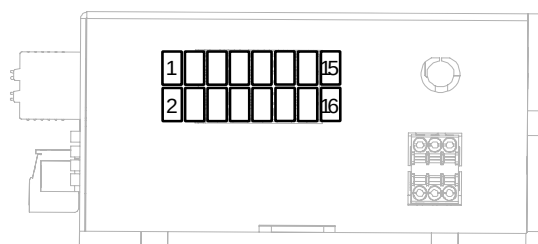
- DIO needs DC24V power input to drive isolated interface.
- Common Contact is used for DI to determine NPN or PNP circuit.

1.5.8. RS485 Connector



Pin	Signal	Pin	Signal
1	GND	2	GND
3	RS485_1 Data+	4	RS485_1 Data-
5	RS485_2 Data+	6	RS485_2 Data-

1.5.9. Encoder Input Connector



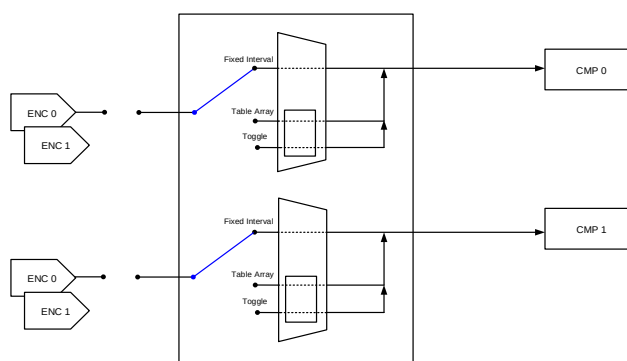
Pin	Label	Description	Pin	Label	Description
1	EA1+	Encoder1 phase A input (+)	2	EA1-	Encoder1 phase A input (-)
3	EB1+	Encoder1 phase B input (+)	4	EB1-	Encoder1 phase B input (-)
5	EA2+	Encoder2 phase A input (+)	6	EA2-	Encoder2 phase A input (-)
7	EB2+	Encoder2 phase B input (+)	8	EB2-	Encoder2 phase B input (-)
9	D5V	DC 5V Output	10	G5V	DC 5V Ground
11	LTC1	Latch Input 1	12	LTC2	Latch Input 2
13	CMP1+	Compare Trigger Out 1 (+)	14	CMP1-	Compare Trigger Out 1 (-)
15	CMP2+	Compare Trigger Out 2 (+)	16	CMP2-	Compare Trigger Out 2 (-)

Note

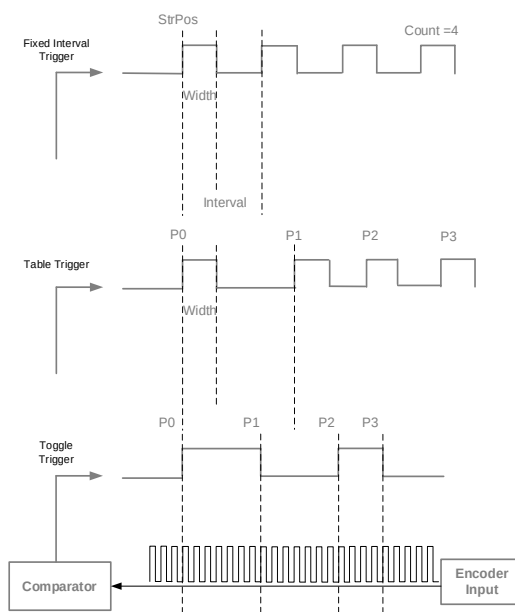
- DC 5V output capacity is 100mA.

ETX-R41K111C is equipped with two comparators (CMP1 ~ CMP2) to catch encoders. Each comparator has its own resources to handle different trigger behaviors. In addition, the comparators can be configured with arbitrary and repeated encoder source. The functions are shown below.

Block Diagram



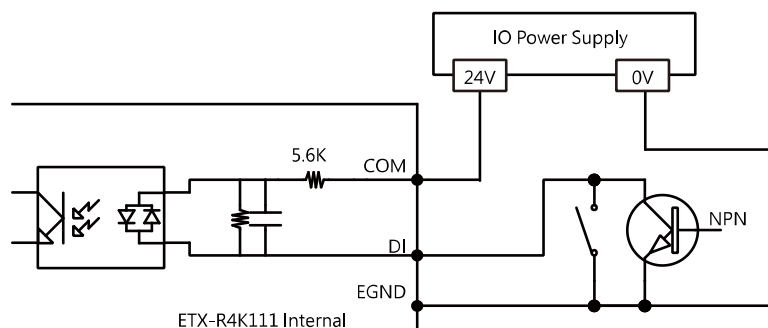
Timing Chart



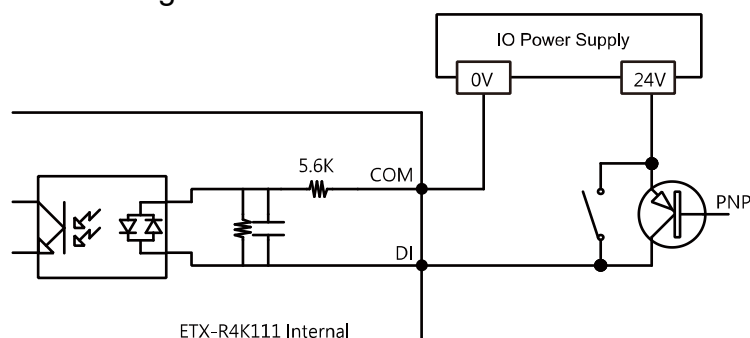
1.6. Signal Circuit

■ Digital Input

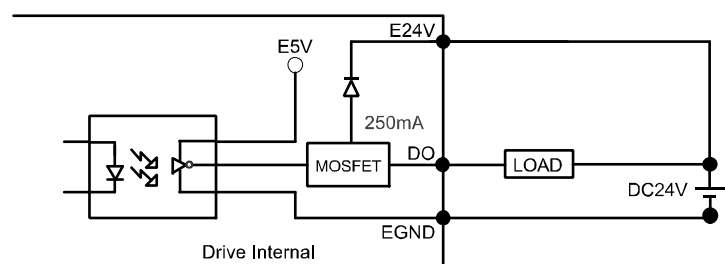
NPN Wiring



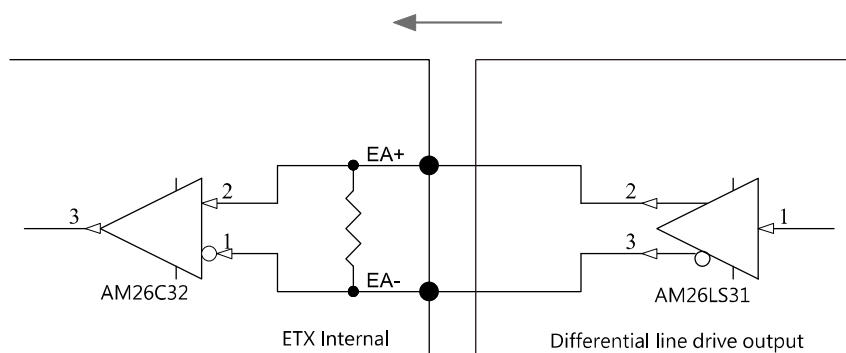
PNP Wiring



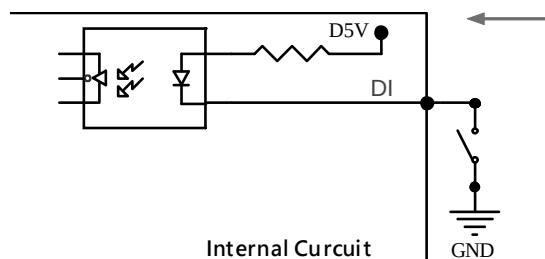
■ Digital Output



Encoder Input

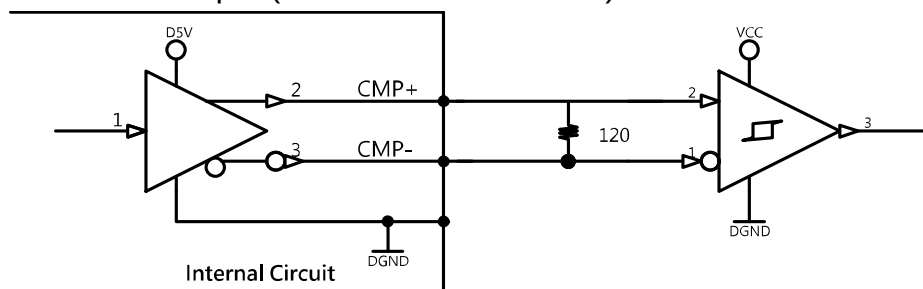


Latch Input Signal Circuit

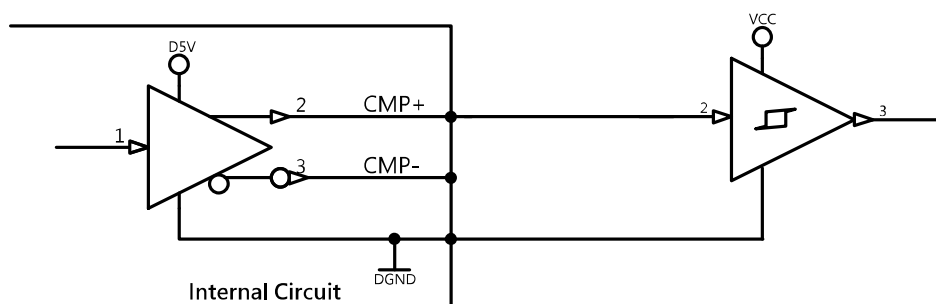


Compare Trigger Output Signal Circuit

Differential output (Line Driver to Receiver)



Differential output (Line Driver to TTL)



Differential Signal Transmission Wiring Notice

When using differential signal transmission, it is important to pay attention to the following key points to ensure signal integrity and reliability:

- Use Twisted Pair Cables**

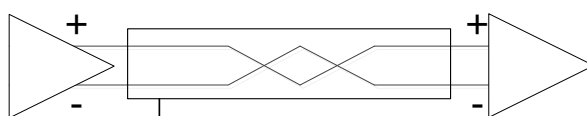
The two wires should be tightly twisted together to ensure that external interference affects both wires equally, allowing the interference to be canceled out at the receiving end.

- Maintain Symmetry**

Ensure that the lengths of the + and - signal lines are the same when wiring to avoid phase differences caused by inconsistent wire lengths.

- Shielding and Grounding**

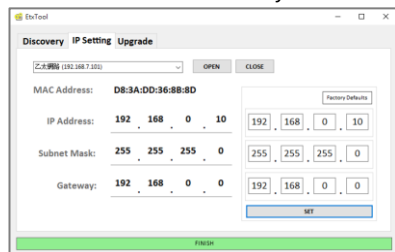
Use shielded twisted pair cables and ground the shielding to further reduce external electromagnetic interference.



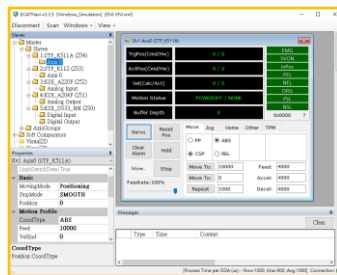
2. Utilities

ETX-R41K111C provides the utilities for users to manage EtherCAT Master and Slaves devices. It provides simple and convenient interface to test the slave devices.

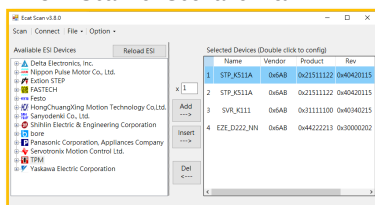
EtXTool for Scenario 2 only



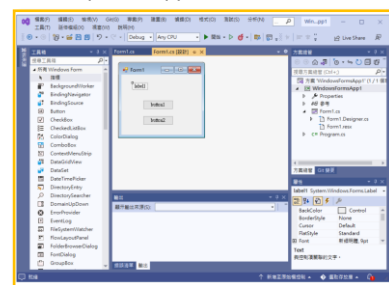
ECATNavi for Scenario 1 & 2



ECATScan for Scenario 1 & 2

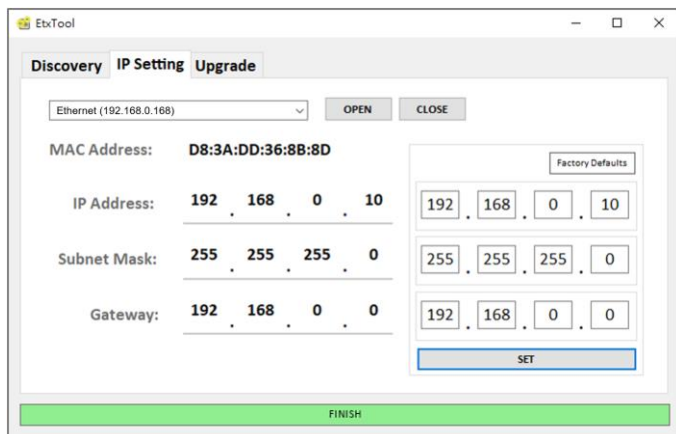


Develop user applications



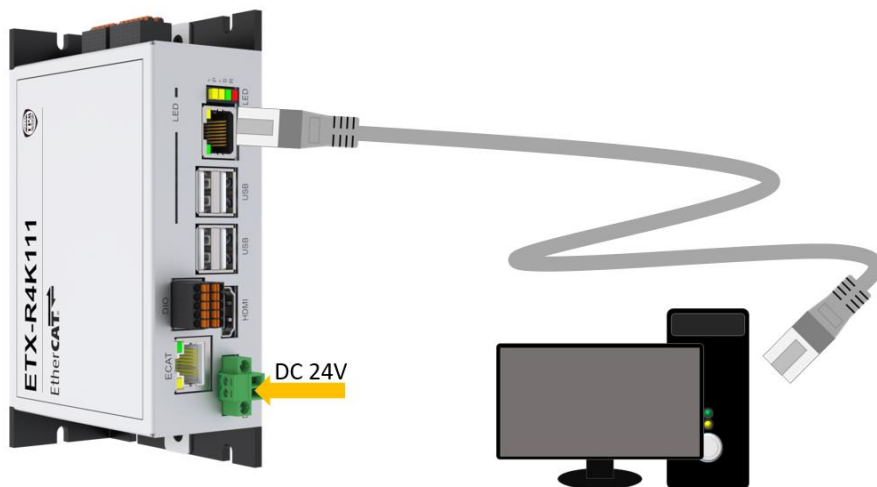
2.1. EtXTool

EtXTool is an initialization tool for operating the ETX-R41K111C in Scenario 2, used when first establishing a connection to the ETX-R41K111C. It helps users configure the IP address and assign it to the ETX-R41K111C, ensuring it matches the network segment of the host PC. Therefore, if configured for Scenario 1, EtXTool is not necessary.

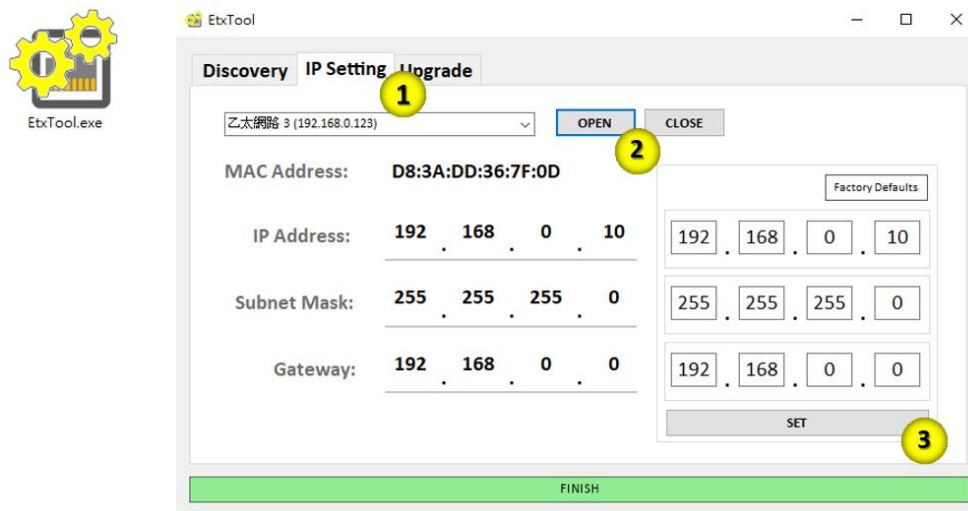


2.1.1. Change IP Settings

Step1: Connect PC's Ethernet port to ETX-R41K111C Ethernet port



Step2: Execute the configuration tool "EtXTool"



1. Select the **IP Setting** tab.
2. Select the PC's Ethernet adapter to which the ETX-R41K111C is connected, and then click the **OPEN** button.
3. Specify the IP Address, Subnet Mask and Gateway of the ETX-R41K111C, and then click the **Set** button to change the configuration.

*Make sure that the ETX-R41K111C is in the same subnet as the PC's Ethernet adapter and that the ETX-R41K111C has a unique IP address.

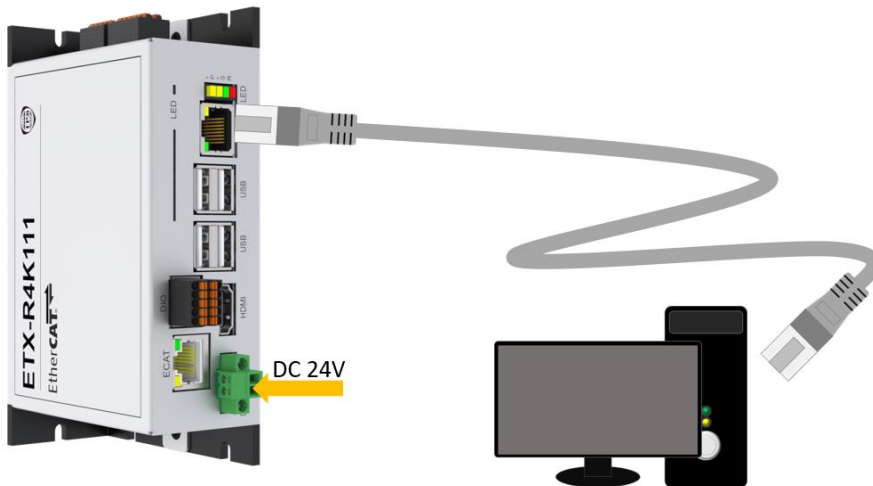
Factory Default

- IP Address: 192.168.0.10
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.0.0

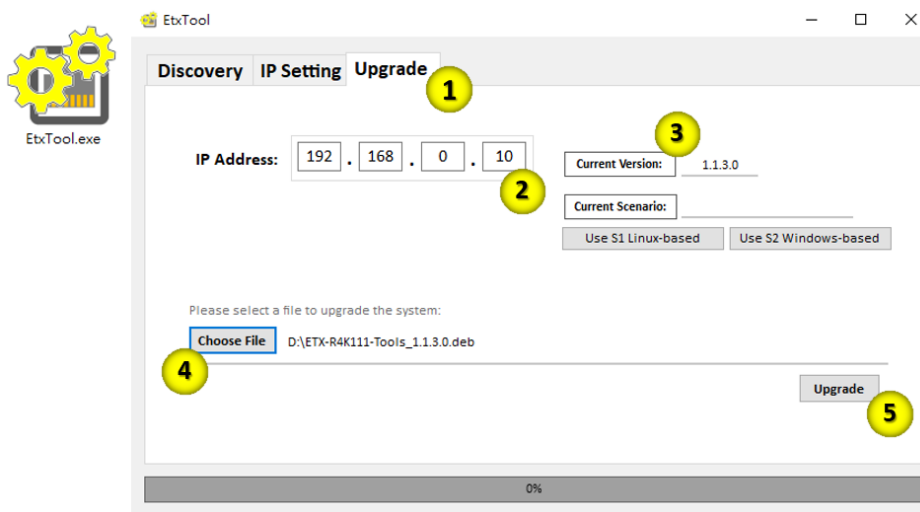
Now, the ETX-R41K111C is using the new configuration.

2.1.2. Upgrade

Step1: Connect PC's Ethernet port to ETX-R41K111C Ethernet port



Step2: Execute the configuration tool "EtXTool"

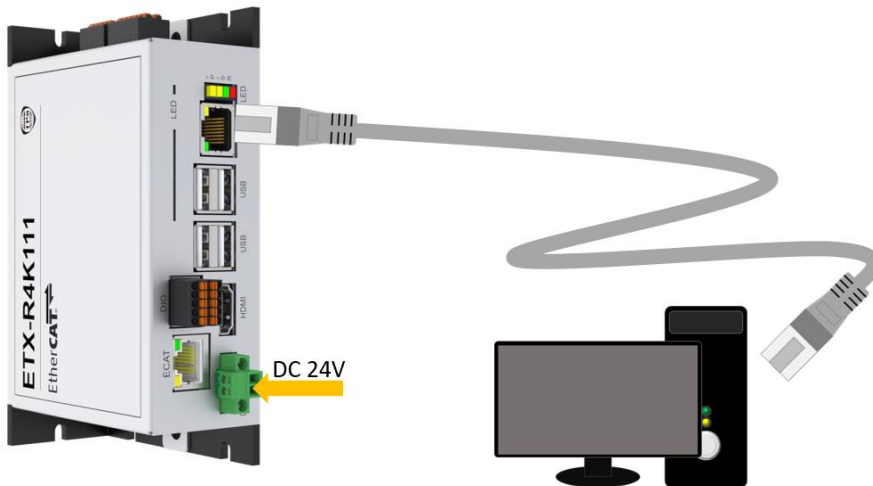


1. Select the **Upgrade** tab.
2. Specify the IP Address of ETX-R41K111C
3. Click the **Current Version** button and the version will be displayed.
4. Click the **Choose File** button and locate the file on your PC.
5. Click the **Upgrade** button.

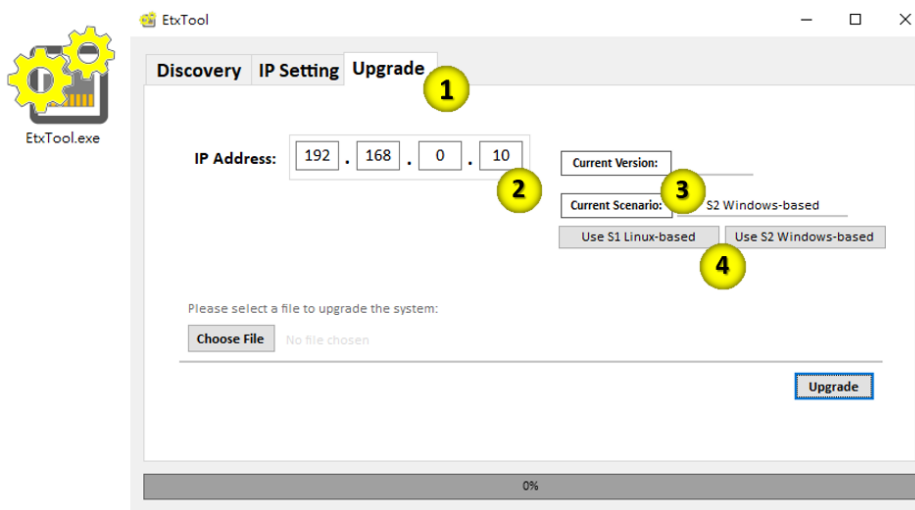
Now, the ETX-R41K111C is using the new software.

2.1.3. Change Scenario

Step1: Connect PC's Ethernet port to ETX-R41K111C Ethernet port



Step2: Execute the configuration tool "EtXTool"



1. Select the **Upgrade** tab.
2. Specify the IP Address of ETX-R41K111C
3. Click the **Current Scenario** button and the current usage scenario will be displayed.
4. Click the **Use S1 Linux-based** or **Use S2 Windows-based** button to change the scenario.

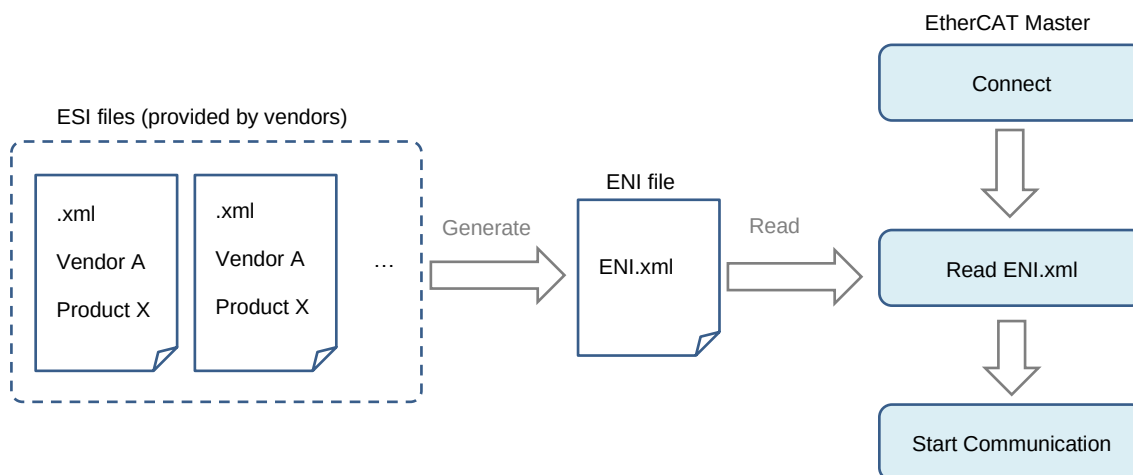
Now, the ETX-R41K111C has been changed to selected scenario.

Note:

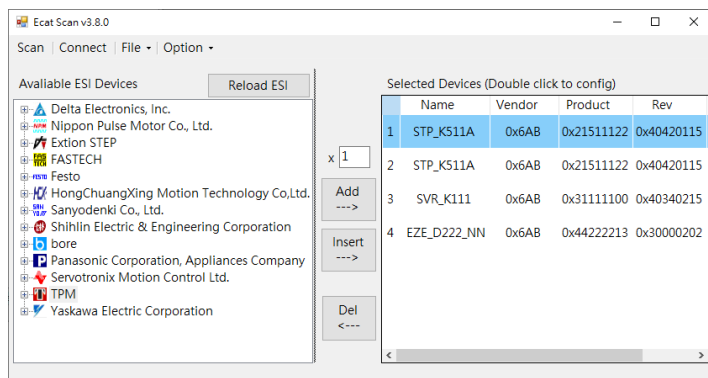
ETX-R41K111C built-in utility "EtXUpgrade" can also change scenario.

2.2. ECATScan

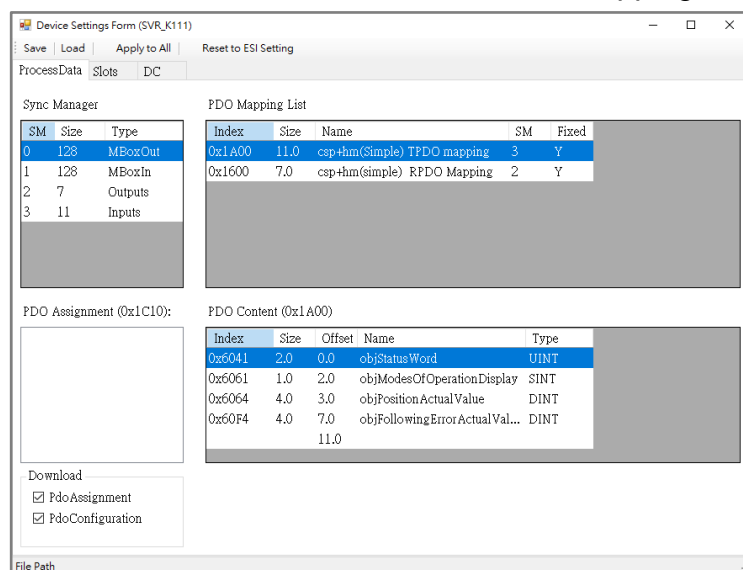
ECATScan is a tool designed to scan, configure, and generate ENI files from ESI files. It has a stand-alone version, ECATScan.exe, and can also be accessed through ECATNavi.exe by clicking the 'Scan' button on the menu bar.



ECATScan can scan the number and types of slave devices on the EtherCAT network.



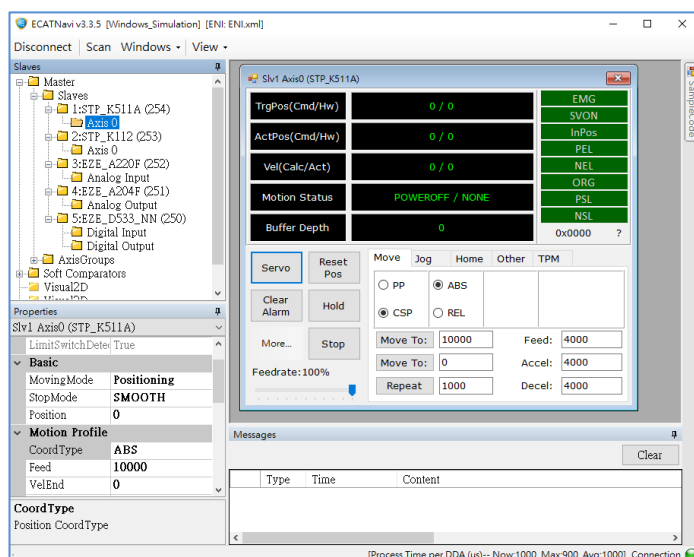
It also allows customization of the PDO mapping for each slave module.



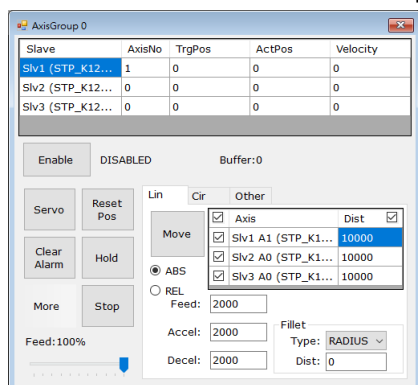
2.3. ECATNavi

ECATNavi is a multifunctional application. It not only tests and manipulates EtherCAT slave devices but also evaluates various functions of ETX-R41K111C, including I/O and the motion library.

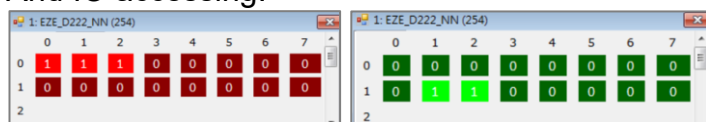
ECATNavi supports various functions, including CSP mode moving, PP mode moving, and Home mode moving.



It also enables multi-axis interpolation by group.



And IO accessing.



3. Project Encryption

This chapter is intended to give a brief overview of the project encryption for ETX-R41K111C. The following section will give background information that is necessary to fully understand the functions and how to encrypt a project of the system.

3.1. Benefits

TPM is a development system provider with ETX-R41K111C as one of the products. System integrators could adapt ETX-R41K111C as the base system to develop applications for their customers. However, customers could find the top source vendor which is TPM and perhaps, the worst case, clone the storage in the system and purchase extra systems from TPM directly. In case of customers bypass the original system provider, which would cut down benefits for the system integrators cooperating with TPM, ETX-R41K111C introduces a technology called project encryption. Through project encryption, the system integrators can lockup certain functionalities or set timers to constrain the system running time. Only the authorized products can be working properly. The authorization is hold by the one and only one system integrator.

With the project encryption technology, the system integrators cooperating with TPM and TPM will be tightly coupled cooperating relationship instead of vicious competition to make a win-win partnership.

3.2. AES Brief Introduction

This standard specifies the Rijndael algorithm, a symmetric block cipher that can process data blocks of 128 bits, using cipher keys with lengths of 128, 192, and 256 bits. Rijndael was designed to handle additional block sizes and key lengths. However they are not adopted in this standard. Throughout the remainder of this standard, the algorithm specified herein will be referred to as “the AES algorithm.” The algorithm may be used with the three different key lengths indicated above, and therefore these different “flavors” may be referred to as “AES-128”, “AES-192”, and “AES-256”.

Since the AES encryption/decryption algorithm is not the main function of ETX-R41K111C, the detailed introduction is not introduced in this manual. Please refer to [Wikipedia](#) for more information.

3.3. Functional Architecture

Each ETX-R41K111C equipped an identification chip with unique serial number. The unique serial number plays the role as the content for AES algorithm. We call the unique serial number the hardware id from now on. Another key held by the system integrator is called the SI key, used for

encryption/decryption for the AES algorithm to make the registration key. The illustration of the making of the registration is as the following figure.



From the above figure, the hardware id is obtainable within ETX-R41K111C, taken as the content for AES algorithm. The SI key, hard coded by the system integrator, is the key to calculate the output value, the registration key. The registration key is like the activation code to activate the full functionalities of the ETX-R41K111C and it is not invertible. Even though the whole data including the registration key could be cloned, the end user or the system integrator's customer cannot obtain the SI key. If an end user wants to buy a replacement from other resources instead of the designated system integrator is not doable since the hardware id would be different with the original one. Therefore, the calculated registration key with the original SI key and different hardware id as content would not match so that the specific functionalities could not be working or the system could only run for certain time period.

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